

Introduction to Coding theory

Ji, Yonghyeon

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We cover the following topics in this note.

- TBA.

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1 Linear Codes

The fundamental problem of coding theory is to transmit information reliably across a noisy channel. A sender begins with a message, usually represented as a vector

$$\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_2^k,$$

and applies an *encoder*

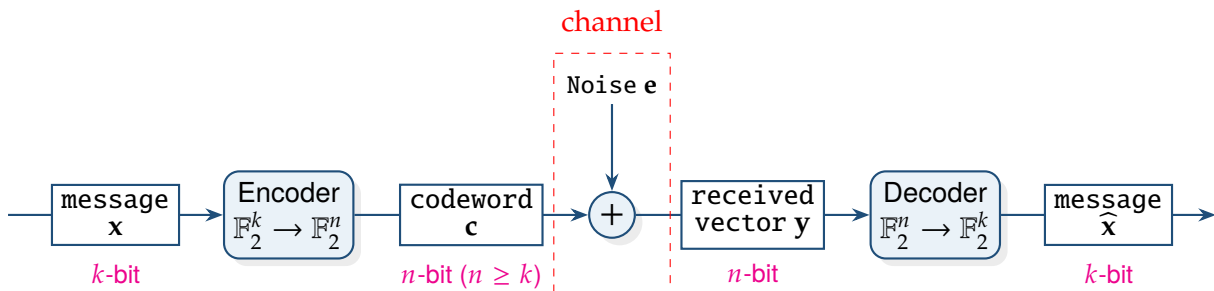
$$\text{Encode} : \mathbb{F}_2^k \longrightarrow \mathbb{F}_2^n, \quad \mathbf{x} \longmapsto \mathbf{c},$$

where $n \geq k$. The vector $\mathbf{c}$ is called the *codeword*.

During transmission, the codeword passes through a noisy channel. The effect of the channel is modeled by adding an error vector $\mathbf{e} \in \mathbb{F}_2^n$ to the transmitted codeword, so that the receiver obtains

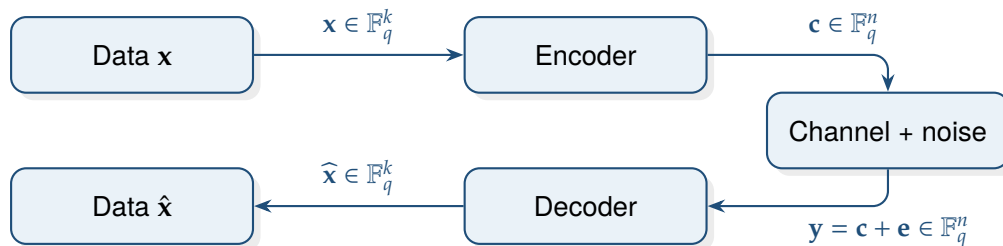
$$\mathbf{y} = \mathbf{c} \oplus \mathbf{e} \in \mathbb{F}_2^n.$$

The vector $\mathbf{y}$ is called the *received vector*. The purpose of the decoder is to recover the original message, or equivalently the transmitted codeword, from the corrupted vector $\mathbf{y}$.



A generic communication model has the form

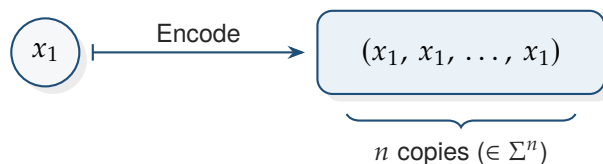
$$\mathbf{x} = (x_1, \dots, x_k) \xrightarrow{\text{encoder}} \underbrace{\mathbf{c} = (c_1, \dots, c_n)}_{\text{codeword } (n \geq k)} \xrightarrow{\text{noise } \mathbf{e}} \mathbf{y} = \mathbf{c} + \mathbf{e} \xrightarrow{\text{decoder}} \hat{\mathbf{x}} = (\hat{x}_1, \dots, \hat{x}_k).$$



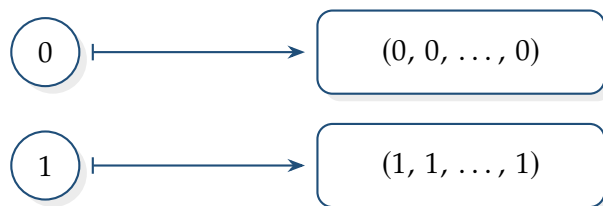
Example 1 (Repetition code). Let Σ be an alphabet. An *encoding* is a map

$$\begin{aligned} \text{Encode} : \quad \Sigma^k &\longrightarrow \Sigma^n \\ (x_1, \dots, x_k) &\longmapsto (c_1, \dots, c_n) \end{aligned}$$

with $n \geq k$.



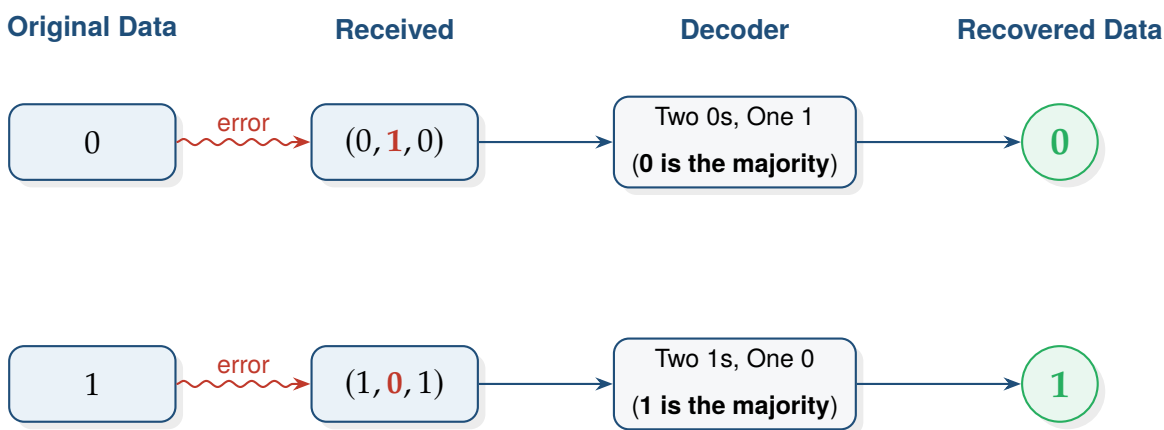
When $k = 1$, a very simple encoding repeats the same symbol



This is called the *repetition code*. If one symbol is changed during transmission, we can still recover the original message by *majority vote*. For instance,

$$(0, 0, 0) \rightsquigarrow (0, 1, 0), \quad (1, 1, 1) \rightsquigarrow (1, 0, 1).$$

Since 0 appears most often in $(0, 1, 0)$, we decode it as 0, and since 1 appears most often in $(1, 0, 1)$, we decode it as 1.



1.1 Definition of Linear Codes

Linear Encoding

Definition 1. A **linear encoding** over $\mathbb{F}_q$ is a linear map

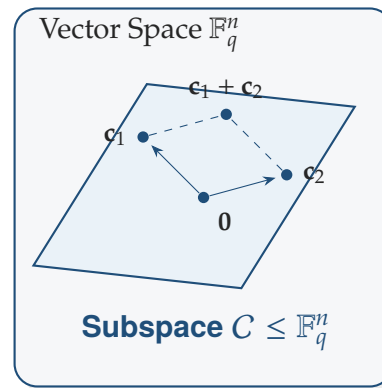
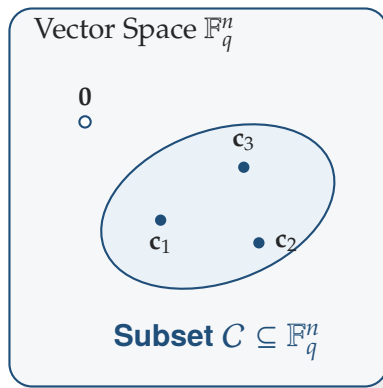
$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \quad \mathbf{x} \longmapsto \mathbf{c}.$$

Linear Code

Definition 2. A **code of length** $n \in \mathbb{Z}_{>0}$ **over** $\mathbb{F}_q$ is a subset

$$C \subseteq \mathbb{F}_q^n.$$

The elements of C are called **codewords**. If $C \leq \mathbb{F}_q^n$, then C is called a **linear code**.



Remark 1. If $\dim_{\mathbb{F}_q}(C) = k$, then C is called a $[n, k]_q$ -**code over** $\mathbb{F}_q$.

Proposition 1. If C is an $[n, k]_q$ -code, then

$$|C| = q^k.$$

Proof. Choose a basis $\mathbf{b}_1, \dots, \mathbf{b}_k$ of C . Every codeword can be written uniquely as

$$x_1 \mathbf{b}_1 + \dots + x_k \mathbf{b}_k, \quad x_i \in \mathbb{F}_q.$$

Each x_i has q choices, and there are k independent coefficients. Therefore the number of codewords is q^k . □

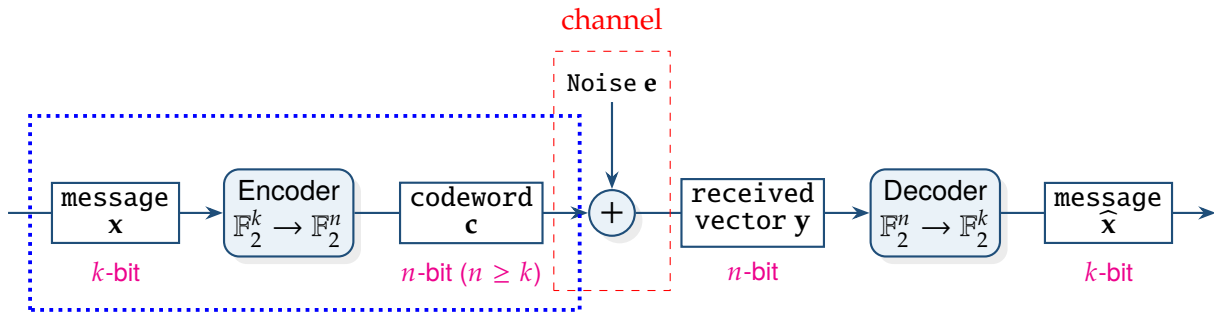
1.2 Generator Matrices

Note. In linear algebra, a choice of bases for finite-dimensional vector spaces V and W identifies each linear transformation $T : V \rightarrow W$ with a matrix.

Accordingly, in coding theory, after choosing bases for $\mathbb{F}_q^k$ and $\mathbb{F}_q^n$, a linear encoding

$$\text{Encode} : \mathbb{F}_q^k \rightarrow \mathbb{F}_q^n$$

is represented by a matrix $\mathbf{G} \in \mathbb{F}_q^{n \times k}$.

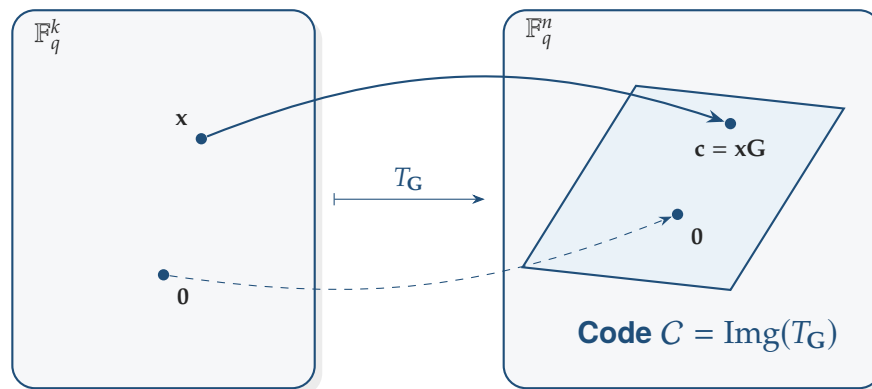


Generator Matrix

Definition 3. Let C be an $[n, k]_q$ -code over $\mathbb{F}_q$. A **generator matrix for C** is a $k \times n$ matrix $\mathbf{G}$ whose rows form a basis of C .

Remark 2. If $\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_q^k$, then encoding is given by

$$\mathbf{c} = \mathbf{x}\mathbf{G} \in \mathbb{F}_q^n \quad \text{with } \mathbf{G} \in \mathbb{F}_q^{k \times n}.$$

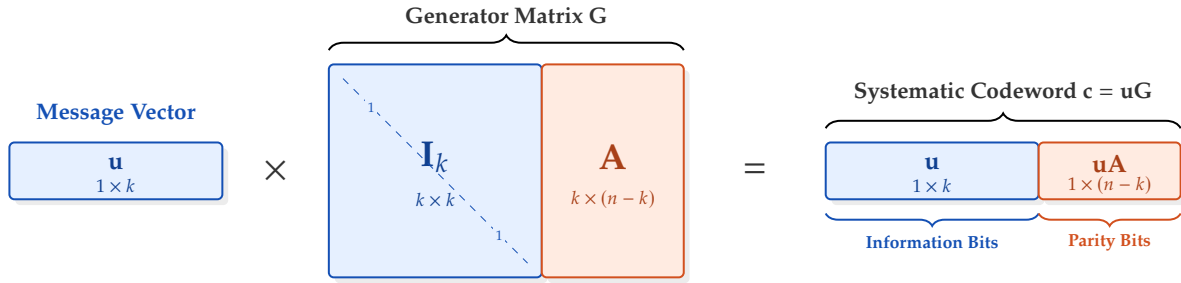


1.2.1 Systematic form

A particularly convenient generator matrix has the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}],$$

where $\mathbf{I}_k$ is the $k \times k$ identity matrix and $\mathbf{A}$ is a $k \times (n - k)$ matrix.



Systematic Form

Definition 4. A code is said to be in **systematic form** if it has a generator matrix of the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$$

Remark 3. In general, a linear code has many generator matrices, since different choices of a basis of the code give different matrices. Thus a generator matrix is not unique.

However, if the code is written in *systematic form* with the first k coordinate positions chosen as the information positions, then there is a unique generator matrix of the form $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$.

Lemma 2. Let $\mathbf{G}, \mathbf{G}' \in \mathbb{F}_q^{k \times n}$ be generator matrices of the same linear code $C \leq \mathbb{F}_q^n$. Then there exists a matrix

$$\exists \mathbf{U} \in GL_k(\mathbb{F}_q) \text{ such that } \mathbf{G}' = \mathbf{UG}.$$

Proof. Let

$$\mathbf{G} = \begin{pmatrix} - & \mathbf{g}_1 & - \\ - & \mathbf{g}_2 & - \\ & \vdots & \\ - & \mathbf{g}_k & - \end{pmatrix}, \quad \mathbf{G}' = \begin{pmatrix} - & \mathbf{g}'_1 & - \\ - & \mathbf{g}'_2 & - \\ & \vdots & \\ - & \mathbf{g}'_k & - \end{pmatrix} \in \mathbb{F}_q^{k \times n}$$

be generator matrices of the same linear code $C \leq \mathbb{F}_q^n$. Then their rows are two bases of the same k -dimensional subspace $C \leq \mathbb{F}_q^n$, so each row of $\mathbf{G}'$ is a linear combination of the rows of $\mathbf{G}$. Thus,

there exist scalars $u_{ij} \in \mathbb{F}_q$ such that

$$\begin{aligned} \mathbf{g}'_1 &= u_{11}\mathbf{g}_1 + u_{12}\mathbf{g}_2 + \cdots + u_{1k}\mathbf{g}_k, \\ \mathbf{g}'_2 &= u_{21}\mathbf{g}_1 + u_{22}\mathbf{g}_2 + \cdots + u_{2k}\mathbf{g}_k, \\ &\vdots \\ \mathbf{g}'_k &= u_{k1}\mathbf{g}_1 + u_{k2}\mathbf{g}_2 + \cdots + u_{kk}\mathbf{g}_k. \end{aligned}$$

Let $\mathbf{U} = (u_{ij}) \in \mathbb{F}_q^{k \times k}$. Then

$$\mathbf{UG} = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1k} \\ u_{21} & u_{22} & \cdots & u_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ u_{k1} & u_{k2} & \cdots & u_{kk} \end{pmatrix} \begin{pmatrix} - & \mathbf{g}_1 & - \\ - & \mathbf{g}_2 & - \\ & \vdots & \\ - & \mathbf{g}_k & - \end{pmatrix} = \begin{pmatrix} - & (u_{11}\mathbf{g}_1 + \cdots + u_{1k}\mathbf{g}_k) & - \\ - & (u_{21}\mathbf{g}_1 + \cdots + u_{2k}\mathbf{g}_k) & - \\ & \vdots & \\ - & (u_{k1}\mathbf{g}_1 + \cdots + u_{kk}\mathbf{g}_k) & - \end{pmatrix} = \begin{pmatrix} - & \mathbf{g}'_1 & - \\ - & \mathbf{g}'_2 & - \\ & \vdots & \\ - & \mathbf{g}'_k & - \end{pmatrix} = \mathbf{G}'.$$

So $\mathbf{G}' = \mathbf{UG}$. Similarly, since the rows of $\mathbf{G}$ are also linear combinations of the rows of $\mathbf{G}'$,

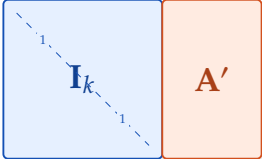
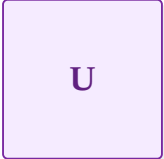
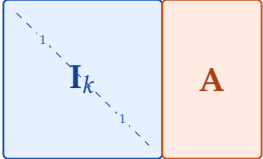
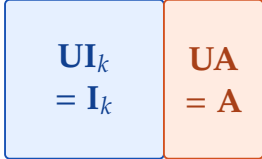
$$\exists \mathbf{V} \in \mathbb{F}_q^{k \times k} \text{ such that } \mathbf{G} = \mathbf{VG}'.$$

Since $\mathbf{G}' = \mathbf{UG}$, we have $\mathbf{G} = \mathbf{VUG}$. Therefore $\mathbf{VU} = \mathbf{I}_k$. Similarly, the equation $\mathbf{G}' = \mathbf{UG} = \mathbf{UVG}$ gives $\mathbf{UV} = \mathbf{I}_k$. Therefore $\mathbf{U}$ is invertible, with inverse $\mathbf{V}$. Hence $\mathbf{U}, \mathbf{V} \in GL_k(\mathbb{F}_q)$. $\square$

Note (Uniqueness of Generator Matrices in Systematic Form). If $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$ and $\mathbf{G}' = [\mathbf{I}_k \mid \mathbf{A}']$ are two generator matrices for the same code, then $\mathbf{G}' = \mathbf{UG}$ for some $\mathbf{U} \in GL_k(\mathbb{F}_q)$. Then

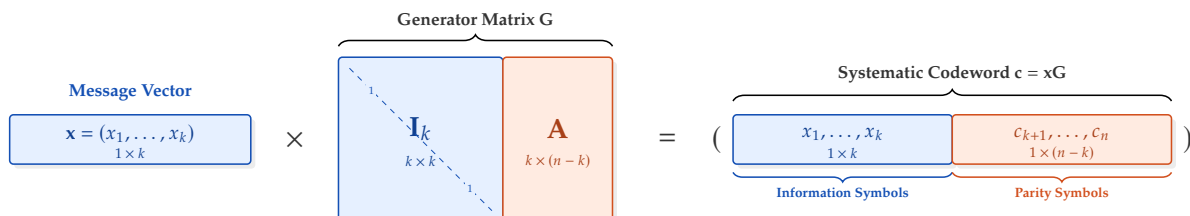
$$\mathbf{G}' = \mathbf{UG} \implies [\mathbf{I}_k \mid \mathbf{A}'] = \mathbf{U}[\mathbf{I}_k \mid \mathbf{A}] \implies [\mathbf{I}_k \mid \mathbf{A}'] = [\mathbf{UI}_k \mid \mathbf{UA}]$$

so $\mathbf{U} = \mathbf{I}_k$. Hence $\mathbf{G}' = \mathbf{G}$, and therefore $\mathbf{A}' = \mathbf{A}$.

Matrix $\mathbf{G}' = [\mathbf{I}_k \mid \mathbf{A}']$	Matrix $\mathbf{U} \in GL_k(\mathbb{F}_q)$	Matrix $\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$	Product $[\mathbf{UI}_k \mid \mathbf{UA}]$
			

Remark 4. Let the first k coordinates of the codeword are exactly the information symbols. If $\mathbf{x} = (x_1, \dots, x_k)$, then

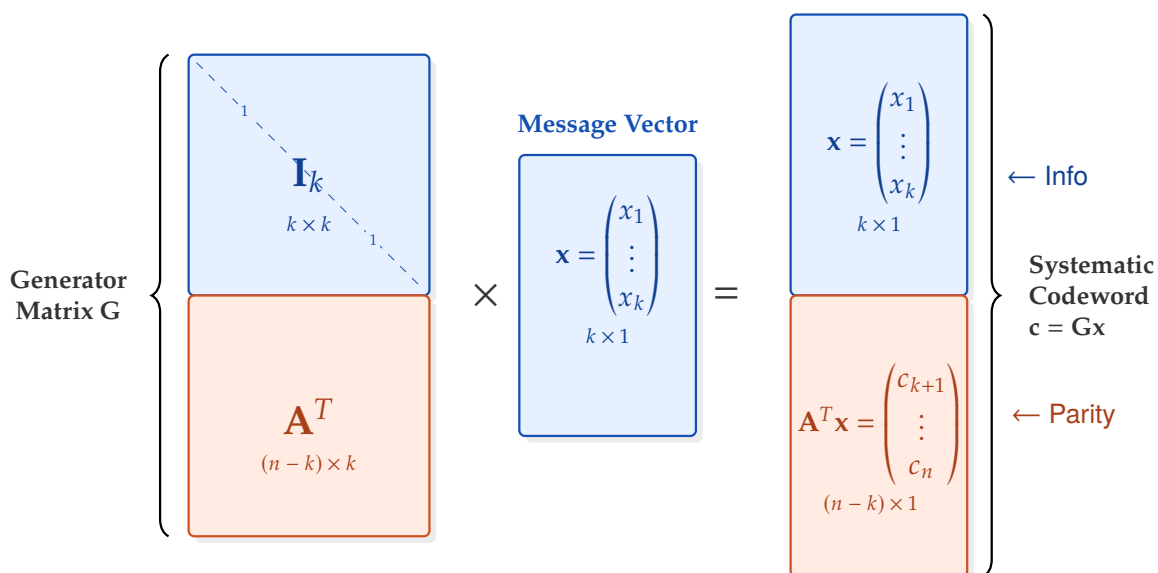
$$\mathbf{xG} = (x_1, \dots, x_k, c_{k+1}, \dots, c_n).$$



So the information symbols appear in the transmitted codeword, followed by the parity symbols.

Note. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix} \in \mathbb{F}_q^k$ be the message vector, written as a column vector. With the column-vector

convention, the encoding map is represented by a generator matrix $\mathbf{G}$ such that $\mathbf{Gx} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \\ c_{k+1} \\ \vdots \\ c_n \end{pmatrix}$.



1.2.2 Examples: Repetition Code and Single Parity-check Code

Example 2 (Repetition code). The $[n, 1]_q$ repetition code is the linear code defined by

$$\text{Encode} : \mathbb{F}_q \longrightarrow \mathbb{F}_q^n, \quad (x_1) \longmapsto (x_1, x_1, \dots, x_1).$$

Its generator matrix is

$$\mathbf{G} = [1 \ 1 \ \dots \ 1]_{1 \times n}.$$

Indeed,

$$(x_1)\mathbf{G} = (x_1)[1 \ 1 \ \dots \ 1] = (x_1, x_1, \dots, x_1).$$

Hence the code is

$$C = \{(a, a, \dots, a) \in \mathbb{F}_q^n : a \in \mathbb{F}_q\} \leq \mathbb{F}_q^n.$$

Example 3 (Single Parity-check Code). The single parity-check code $[n, n-1]_q$ is the linear code

$$C = \left\{ (x_1, \dots, x_{n-1}, \sum_{i=1}^{n-1} x_i) \in \mathbb{F}_q^n : (x_1, \dots, x_k) \in \mathbb{F}_q^k \right\} \leq \mathbb{F}_q^n.$$

Equivalently, it is defined by the encoding map

$$\text{Encode} : \mathbb{F}_q^{n-1} \longrightarrow \mathbb{F}_q^n, \quad (x_1, \dots, x_{n-1}) \longmapsto (x_1, \dots, x_k, \sum_{i=1}^{n-1} x_i).$$

A generator matrix in systematic form is

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & \dots & 0 & 1 \\ 0 & 1 & \dots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 1 \end{bmatrix}_{k \times n} = \begin{bmatrix} & 1 \\ \mathbf{I}_k & \vdots \\ & 1 \end{bmatrix} = [\mathbf{I}_k \mid \mathbf{1}],$$

where $\mathbf{1} \in \mathbb{F}_q^k$ denotes the column vector all of whose entries are 1. Indeed, for $\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{F}_q^k$,

$$\mathbf{x}\mathbf{G} = (x_1, \dots, x_k) \begin{bmatrix} & 1 \\ \mathbf{I}_k & \vdots \\ & 1 \end{bmatrix} = (x_1, \dots, x_k, x_1 + \dots + x_k).$$

Hence the first $n-1$ coordinates are the information symbols, and the last coordinate is their parity symbol.

1.3 Parity-check Matrices

A linear code can be described either by its generators or by the linear equations that all codewords satisfy.

Parity-check Matrix

Definition 5. Let C be an $[n, k]_q$ -code. A matrix $H \in \mathbb{F}_q^{(n-k) \times n}$ is called a **parity-check matrix** for C if

$$C = \{\mathbf{v} \in \mathbb{F}_q^n : H\mathbf{v}^T = 0\}.$$

Thus codewords are exactly the vectors satisfying a system of $n - k$ homogeneous linear equations.

1.3.1 Parity-check Matrix from a Systematic Generator Matrix

Note. Suppose

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}],$$

is a generator matrix in systematic form, where $\mathbf{A} \in \mathbb{F}_q^{k \times (n-k)}$. Then define

$$\mathbf{H} = [-\mathbf{A}^T \mid \mathbf{I}_{n-k}].$$

Parity-check Matrix of a Systematic Generator Matrix

Proposition 3. Let C be an $[n, k]_q$ -code, and suppose that C is generated by the systematic generator matrix

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}] \in \mathbb{F}_q^{k \times n},$$

where $\mathbf{A} \in \mathbb{F}_q^{k \times (n-k)}$. Then

$$\mathbf{H} = [-\mathbf{A}^T \mid \mathbf{I}_{n-k}] \in \mathbb{F}_q^{(n-k) \times n}$$

is a parity-check matrix of C .

Proposition 4 (Parity-check matrix of a systematic generator matrix). Let C be an $[n, k]_q$ -code, and suppose that C is generated by the systematic generator matrix

$$G = [I_k \mid A] \in M_{k \times n}(\mathbb{F}_q),$$

where $A \in M_{k \times (n-k)}(\mathbb{F}_q)$. Then

$$H = [-A^T \mid I_{n-k}] \in M_{(n-k) \times n}(\mathbb{F}_q)$$

is a parity-check matrix of C .

Proof. By definition, it suffices to show that

$$GH^{\top} = 0.$$

Now

$$H^{\top} = \begin{bmatrix} -A \\ I_{n-k} \end{bmatrix},$$

so

$$GH^{\top} = [I_k \mid A] \begin{bmatrix} -A \\ I_{n-k} \end{bmatrix} = I_k(-A) + AI_{n-k} = -A + A = 0.$$

□

Note (Linear Code as Short Exact Sequence). A linear $[n, k]_q$ code is most naturally described by a short exact sequence

$$0 \longrightarrow \mathbb{F}_q^k \xrightarrow{\text{Encode}} \mathbb{F}_q^n \xrightarrow{H} \mathbb{F}_q^{n-k} \longrightarrow 0,$$

A message $\mathbf{x} \in \mathbb{F}_q^k$ is encoded as

$$\mathbf{c} = \text{Enc}(\mathbf{x}) \in C.$$

After transmission through the channel, one receives

$$\mathbf{y} = \mathbf{c} + \mathbf{e} \in \mathbb{F}_q^n,$$

where $\mathbf{e} \in \mathbb{F}_q^n$ is the error vector. Applying the parity-check map yields

$$H(\mathbf{y}) = H(\mathbf{c} + \mathbf{e}) = H(\mathbf{e}),$$

since $H(\mathbf{c}) = 0$ for every $\mathbf{c} \in C$.

Thus decoding may be viewed as follows: from the syndrome $H(\mathbf{y})$, choose an error estimate $\hat{\mathbf{e}}$ with the same syndrome, typically of smallest Hamming weight, and define

$$\hat{\mathbf{c}} = \mathbf{y} - \hat{\mathbf{e}} \in C.$$

Then recover the message by

$$\hat{\mathbf{x}} = \text{Enc}^{-1}(\hat{\mathbf{c}}).$$

Hence error correction is the procedure of moving the received word from its ambient coset in $\mathbb{F}_q^n/C$ back to the code $C = \ker(H)$, and then inverting the encoding map.

Cardinality of the matrix space $\text{Mat}_m(\mathbb{F}_q)$

Proposition 5. Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of $m \times n$ matrices over the finite field $\mathbb{F}_q$ is

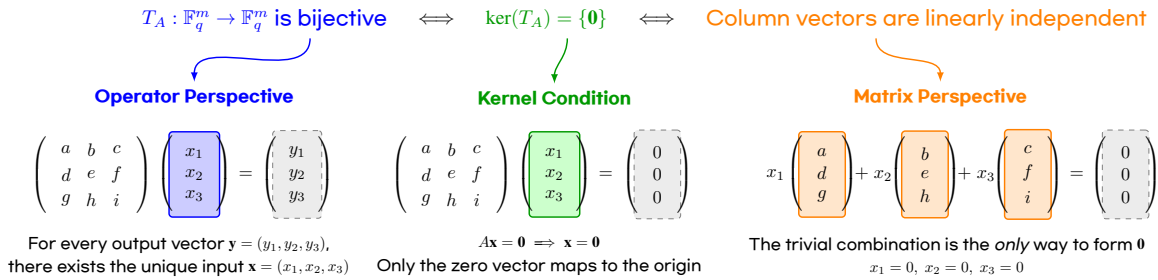
$$|\text{Mat}_{m \times n}(\mathbb{F}_q)| = q^{mn}.$$

In particular, $|\text{Mat}_m(\mathbb{F}_q)| = q^{m^2}$.

Remark 5. Let

$$A = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_m \\ | & | & \cdots & | \end{pmatrix} \in \text{Mat}_{m \times m}(\mathbb{F}_q),$$

where $\mathbf{v}_i \in \mathbb{F}_q^m$ are the columns of A . Then A is invertible if and only if its columns are linearly independent.



Observation (Computation of $|\text{GL}_2(\mathbb{F}_q)|$).

A 2×2 matrix has the form

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad a, b, c, d \in \mathbb{F}_q.$$

Write the columns as

$$\mathbf{v}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} b \\ d \end{pmatrix}.$$

$$|\text{GL}_2(\mathbb{F}_q)| = (q^2 - 1) \cdot (q^2 - q)$$

$$\begin{pmatrix} \boxed{a} & b \\ \boxed{c} & d \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & \boxed{b} \\ \boxed{c} & \boxed{d} \end{pmatrix}$$

$$\begin{matrix} \mathbf{v}_1 \neq \mathbf{0} \\ q^2 - 1 \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_2 \notin \text{Span}(\mathbf{v}_1) \\ q^2 - q \text{ ways} \end{matrix}$$

(i) The vector $\mathbf{v}_1$ can be any nonzero vector in $\mathbb{F}_q^2$. Hence there are $q^2 - 1$ choices.

(ii) Since $\text{span}(\mathbf{v}_1) = \{\lambda \mathbf{v}_1 : \lambda \in \mathbb{F}_q\}$ has exactly q elements, there are $q^2 - q$ choices for $\mathbf{v}_2$.

Multiplying these choices, we obtain

$$|\text{GL}_2(\mathbb{F}_q)| = (q^2 - 1)(q^2 - q).$$

Observation (Computation of $|GL_3(\mathbb{F}_q)|$).

$$|GL_3(\mathbb{F}_q)| = (q^3 - 1) \cdot (q^3 - q) \cdot (q^3 - q^2)$$

$$\begin{pmatrix} \boxed{a} & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & \boxed{b} & c \\ d & e & f \\ g & h & i \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & b & \boxed{c} \\ d & e & f \\ g & h & i \end{pmatrix}$$

$$\begin{matrix} \mathbf{v}_1 \neq \mathbf{0} \\ q^3 - 1 \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_2 \notin \text{Span}(\mathbf{v}_1) \\ q^3 - q \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_3 \notin \text{Span}(\mathbf{v}_1, \mathbf{v}_2) \\ q^3 - q^2 \text{ ways} \end{matrix}$$

Observation (Computation of $|GL_4(\mathbb{F}_q)|$).

$$|GL_4(\mathbb{F}_q)| = (q^4 - 1) \cdot (q^4 - q) \cdot (q^4 - q^2) \cdot (q^4 - q^3)$$

$$\begin{pmatrix} \boxed{a} & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & \boxed{b} & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & b & \boxed{c} & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix} \quad \begin{pmatrix} \boxed{a} & b & c & \boxed{d} \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$$

$$\begin{matrix} \mathbf{v}_1 \neq \mathbf{0} \\ q^4 - 1 \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_2 \notin \text{Span}(\mathbf{v}_1) \\ q^4 - q \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_3 \notin \text{Span}(\mathbf{v}_1, \mathbf{v}_2) \\ q^4 - q^2 \text{ ways} \end{matrix} \quad \times \quad \begin{matrix} \mathbf{v}_4 \notin \text{Span}(\mathbf{v}_i)_{i=1}^3 \\ q^4 - q^3 \text{ ways} \end{matrix}$$

Counting invertible matrices over a finite field $\mathbb{F}_q$

Theorem 6. Let $m, n \in \mathbb{Z}_{>0}$, and let q be a prime power. The number of invertible $m \times m$ matrices over $\mathbb{F}_q$ is

$$|GL_m(\mathbb{F}_q)| = (q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1}).$$

Equivalently, $|GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i).$

Remark 6. Recall that

$$|\text{Mat}_m(\mathbb{F}_q)| = q^{m^2}, \quad \text{and} \quad |GL_m(\mathbb{F}_q)| = \prod_{i=0}^{m-1} (q^m - q^i),$$

and hence

$$\begin{aligned}
 \frac{|GL_m(\mathbb{F}_q)|}{|\text{Mat}_m(\mathbb{F}_q)|} &= \frac{\prod_{i=0}^{m-1} (q^m - q^i)}{q^{m^2}} = \frac{1}{q^{m^2}} \left[(q^m - 1)(q^m - q)(q^m - q^2) \cdots (q^m - q^{m-1}) \right] \\
 &= \frac{1}{q^{m^2}} \left[q^m \left(1 - \frac{1}{q^m}\right) \left(1 - \frac{1}{q^{m-1}}\right) \left(1 - \frac{1}{q^{m-2}}\right) \cdots \left(1 - \frac{1}{q^1}\right) \right] \\
 &= \left(1 - \frac{1}{q^m}\right) \left(1 - \frac{1}{q^{m-1}}\right) \left(1 - \frac{1}{q^{m-2}}\right) \cdots \left(1 - \frac{1}{q^1}\right) \\
 &= \prod_{j=1}^m (1 - q^{-j}).
 \end{aligned}$$

m	$ \text{Mat}_m(\mathbb{F}_q) $	$ GL_m(\mathbb{F}_q) $	$\frac{ GL_m(\mathbb{F}_q) }{ \text{Mat}_m(\mathbb{F}_q) }$
1	q	$q - 1$	$1 - q^{-1}$
2	q^4	$(q^2 - 1)(q^2 - q)$	$(1 - q^{-1})(1 - q^{-2})$
3	q^9	$(q^3 - 1)(q^3 - q)(q^3 - q^2)$	$(1 - q^{-1})(1 - q^{-2})(1 - q^{-3})$
4	q^{16}	$(q^4 - 1)(q^4 - q)(q^4 - q^2)(q^4 - q^3)$	$(1 - q^{-1})(1 - q^{-2})(1 - q^{-3})(1 - q^{-4})$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
m	q^{m^2}	$\prod_{i=0}^{m-1} (q^m - q^i)$	$\prod_{j=1}^m (1 - q^{-j})$

Example 4 (Invertible binary matrices). For $q = 2$, we have

$$|\text{Mat}_m(\mathbb{F}_2)| = 2^{m^2}, \quad \frac{|GL_m(\mathbb{F}_2)|}{|\text{Mat}_m(\mathbb{F}_2)|} = \prod_{j=1}^m (1 - 2^{-j}).$$

m	$ \text{Mat}_m(\mathbb{F}_2) $	$ GL_m(\mathbb{F}_2) $	$\frac{ GL_m(\mathbb{F}_2) }{ \text{Mat}_m(\mathbb{F}_2) }$
1	2	1	0.500000
2	16	6	0.375000
3	512	168	0.328125
4	65,536	20,160	0.307617
$\vdots$	$\vdots$	$\vdots$	$\vdots$
20	—	—	0.288788

```

q = 2

print(f"{'m':>2} {'|Mat_m(F_q)|':>15} {'|GL_m(F_q)|':>15} {'ratio':>12}")
print("-" * 48)

for m in range(1, 7):
    Mat = q^(m^2)
    GL = prod(q^m - q^i for i in range(m))
    ratio = GL / Mat
    print(f"{'m':>2} {'Mat':>15} {'GL':>15} {'ratio.n():>12.6f}")

```

m	Mat_m(F_q)	GL_m(F_q)	ratio
1	2	1	0.500000
2	16	6	0.375000
3	512	168	0.328125
4	65536	20160	0.307617
5	33554432	9999360	0.298004
6	68719476736	20158709760	0.293348

Proposition 7. Let $q \geq 2$ be fixed, and define

$$a_m := \frac{|GL_m(\mathbb{F}_q)|}{|\text{Mat}_m(\mathbb{F}_q)|} = \prod_{j=1}^m (1 - q^{-j}).$$

Then the sequence $\langle a_m \rangle_{m \geq 1}$ converges as $m \rightarrow \infty$. In other words, there exists $\lim_{n \rightarrow \infty} \prod_{j=1}^m (1 - q^{-j}) \in \mathbb{R}$.

Proof. For every $j \geq 1$,

$$q \geq 2 \implies q^j \geq 2^j > 1 \implies 0 < \frac{1}{q^j} < 1.$$

Therefore

$$0 < \frac{1}{q^j} < 1 \implies -1 < -\frac{1}{q^j} < 0 \implies 0 < 1 - \frac{1}{q^j} < 1.$$

It follows that for each $m \geq 1$,

$$a_m = \prod_{j=1}^m (1 - q^{-j}) > 0.$$

So the sequence $\langle a_m \rangle$ is bounded below by 0. Moreover,

$$a_{m+1} = \prod_{j=1}^{m+1} (1 - q^{-j}) = \left[\prod_{j=1}^m (1 - q^{-j}) \right] \cdot (1 - q^{-(m+1)}) = a_m (1 - q^{-(m+1)}).$$

Since $0 < 1 - q^{-(m+1)} < 1$, we obtain $a_{m+1} < a_m$. Thus $\langle a_m \rangle$ is decreasing. Therefore $\langle a_m \rangle$ is a monotone decreasing sequence which is bounded below. Hence it converges. $\square$

Example 5. For small prime q , the limit is approximately

$$\begin{aligned} q = 2 : & \quad \prod_{j=1}^{\infty} (1 - 2^{-j}) \approx 0.288788095 \\ q = 3 : & \quad \prod_{j=1}^{\infty} (1 - 3^{-j}) \approx 0.560126077 \\ q = 5 : & \quad \prod_{j=1}^{\infty} (1 - 5^{-j}) \approx 0.760332795 \end{aligned}$$

```
q = 2
r = 3
s = 5
R = RealField(100)

L = prod(R(1) - R(q)^(-j) for j in range(1, 100))
M = prod(R(1) - R(r)^(-j) for j in range(1, 100))
N = prod(R(1) - R(s)^(-j) for j in range(1, 100))

print(L)
print(M)
print(N)
```

```
0.28878809508660242127889972193
0.56012607792794894496979224331
0.76033279587123242010148829629
```

1.4 Weights and Distances

2 Cyclic Codes

2.1 Historical Development of Cyclic Codes

Year	Milestone	Description
1948	Birth of Coding Theory	Shannon established information theory and proved the existence of reliable codes [1].
1950	Hamming Codes	Hamming introduced the first systematic single-error-correcting linear codes [2].
1954	Reed–Muller Codes	Reed and Muller introduced multiple-error-correcting codes from Boolean functions [3][4].
1957	Algebraic Foundation of Cyclic Codes	Prange identified cyclic codes with ideals in $\mathbb{F}_q[X]/\langle X^n - 1 \rangle$ [5].
1959–1960	BCH Codes	Hocquenghem, Bose, and Ray-Chaudhuri developed BCH codes with designed distance [6][7].
1960	Reed–Solomon Codes	Reed and Solomon introduced powerful nonbinary cyclic codes [8].
1961	Peterson’s Decoding Algorithm	Peterson developed algebraic BCH decoding and published a foundational textbook [9].
1968	Berlekamp–Massey Algorithm	Berlekamp and Massey introduced an efficient error-locator algorithm [10][11].
1970s	Industrial Standardization of CRC	CRC became standard because cyclic codes allow efficient LFSR implementation.
1977	Goppa Codes and the McEliece Cryptosystem	Goppa introduced algebraic geometry codes, and McEliece applied them to cryptography [12][13].
Since 1990s	Modern Developments	Cyclic codes remained important despite the rise of turbo and LDPC codes [14][15].

2.2 What Is a Cyclic Code?

Note. A cyclic code is a special subclass of linear codes that satisfies two key properties

- (i) **linearity** and
- (ii) **invariance under cyclic shift.**

Cyclic code

Definition 6. A linear code C of length n is called a cyclic code if it satisfies both of the following

- (i) **Linearity:** for any two codewords $\mathbf{c}_1, \mathbf{c}_2 \in C$, we have $\mathbf{c}_1 + \mathbf{c}_2 \in C$.
- (ii) **Cyclic shift property:** if $\mathbf{c} = (c_0, c_1, \dots, c_{n-1}) \in C$, then the vector

$$\mathbf{c}^{\gg 1} = (c_{n-1}, c_0, c_1, \dots, c_{n-2})$$

must also belong to C .

Example 6 (Binary $[7, 4]_2$ cyclic code). Suppose

$$\mathbf{c} = (1, 0, 0, 1, 0, 1, 1)$$

is a codeword. Then

- $\mathbf{c}^{\gg 1} = (1, 1, 0, 0, 1, 0, 1)$
- $\mathbf{c}^{\gg 2} = (1, 1, 1, 0, 0, 1, 0)$
- $\mathbf{c}^{\gg 3} = (0, 1, 1, 1, 0, 0, 1)$

All of these vectors must belong to the code space C . Let $n = 4$ and consider the set

$$C = \{(0, 0, 0, 0), (1, 0, 1, 0), (0, 1, 0, 1), (1, 1, 1, 1)\}.$$

- Linearity check: $(1, 0, 1, 0) \oplus (0, 1, 0, 1) = (1, 1, 1, 1) \in C$.
- Cyclic-shift check:
 - $(1, 0, 1, 0)$ shifted once gives $(0, 1, 0, 1) \in C$.
 - $(0, 1, 0, 1)$ shifted once gives $(1, 0, 1, 0) \in C$.
 - $(1, 1, 1, 1)$ shifted once gives $(1, 1, 1, 1) \in C$.

Hence this is a cyclic code.

2.2.1 Example 1.3: A Linear Code That Is Not Cyclic

Let

$$C' = \{(0, 0, 0, 0), (1, 1, 0, 0), (0, 0, 1, 1), (1, 1, 1, 1)\}.$$

Shifting $(1, 1, 0, 0)$ one step cyclically gives $(0, 1, 1, 0)$, which is not in C' . Therefore C' is linear, but **not** cyclic.

A Exercises

References

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